[redacted]
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Statement of Purpose | Ph.D. applicant, Fall 2024
Electrical & Computer Engineering @ [redacted]

Motivation. As we shrink transistors to atomic sizes, fundamental limitations challenge traditional transistor scaling trends. Over the last decade, these trends have witnessed a significant decline, necessitating us to rethink our algorithms and design application-specific accelerators to make new technologies faster and more usable. This requirement is particularly critical for the demanding computational needs of modern cryptographic protocols. Motivated along these lines, my interests in security, embedded systems, and computer architecture have drawn me in the following directions:

- Processors and Hardware Accelerators: How to design scalable hardware architectures for cryptographic algorithms? Specifically Homomorphic Encryption for Privacy-Preserving applications.
- Embedded Systems Security: How to design energy-efficient end devices with cryptographic primitives for ensuring data privacy in this rapidly growing IoT ecosystem?

Finding My Research Interests. My research journey began under [redacted] during the summer of 2022 at IIT Bhilai. We designed a covert communication channel superposed over LoRa transmissions and its corresponding receiver algorithm. While covert transmission over LoRa packets has been studied widely, only few analyze the possibility of their channel staying undetected by the network's server. Our work addressed some of these limitations and presented an alternate design – *CoLoRa*. My contribution was developing a fully parameterized software simulator to analyze the performance of *CoLoRa* under various conditions. This work led to a recently accepted paper at *National Conference on Communications 2024*. Though the paper had previously experienced a few submissions, which was initially disheartening, I can recognize how much the overall quality of the paper has improved after each review. Through this project, I learned extensively about the vulnerabilities associated with low-resource IoT devices and experienced the dynamics of research work, which I thoroughly enjoyed.

Following the summer experience, I was keen to explore multiple research areas, which was promptly fulfilled by the opportunities provided by my research fellowship. I continued as [redacted]'s advisee, and in collaboration with [redacted] at IIT Delhi, I worked on various problem statements that lay at the intersection of Robotics and Networking with applications of ML. Most of my contributions were towards two activities – 1) Demonstration of a distributed architecture of ML models to initiate corrective actions at the edge, and 2) Lightweight shape detection models for the Allegro Robotic Hand (ARH). For the first activity, we demonstrated a cooperating set of three ML algorithms executing on edge nodes to initiate corrective actions intended to control repercussions that arise due to latency in communication. We showed the existence of a general family of such algorithms with tunable parameters chosen as per requirements. This project led to my first journal submission and is presently in the second round of revisions at *Wireless Personal Communications*. The next activity was a comprehensive study of the forces experienced and angles configured at the ARH when holding different objects. Working on this rich data often requires intense computing power, which can be arduous on edge nodes. In response to this need, our study exploited the high correlation among several finger joints and devised an efficient latent representation to classify thirteen shapes. This paper is in the second round of revisions at *IEEE Transactions on Haptics*. In both activities, I was actively involved from the problem formulation phase up until the end of the manuscript writing activity. I wrote all programs for simulations, performed experiments, and consulted with other researchers in this area. These projects fall under a broader umbrella of efforts led by the Department of Telecommunications (DoT), India, towards developing a Mobile Edge Computing (MEC) system for the indigenous 5G test-bed; working on such projects of national interest made me realize the impact of my work and also gave me opportunities to interact with experienced professionals.

Parallel to the fellowship activities and in pursuit of finding my research interests, I took up a few advanced courses in my third year, juggling the coursework, and my projects were at times hectic, which impacted my GPA. Nevertheless, this coalescence of rigorous coursework and fellowship discussions played a pivotal role in finding my forte. The repeated emphasis on the need for lightweight algorithms for fast inference and addressing resource constraints of edge nodes during my fellowship discussions made me

ponder architectural innovations specific to algorithms. My first project (*CoLoRa*) had entrenched notions of security in mind, which intensified through courses and several engagements with [redacted]. From coursework, I was fortunate to get hands-on experience with FPGAs and embedded systems, focussing on applications along the lines of security. However, the limited scope of research in this intersection at my undergraduate institute made me seek external internships to explore my newfound interests.

With the above interests in heart, I spent the summer of 2023 at IISc Bangalore as a research intern, jointly advised by [redacted] and [redacted] (IBM Research); we worked on an exciting intersection of hardware and Zero-Knowledge Proofs (ZKP). I conducted a thorough study on a widely used zkSNARK – PLONK, searching for opportunities for hardware acceleration specific to this protocol rather than zkSNARKs in general. My main contributions during this period were identifying compute-intensive operations in each prover protocol step and estimating hardware requirements. One fundamental target of acceleration was the Number Theoretic Transform (NTT) operation over large field sizes and polynomial degrees, and this realization opened me to a plethora of ongoing research at the intersection of cryptography and hardware (centered around Homomorphic Encryption and Post-Quantum Cryptography) that rely on efficient parameter-scalable implementations of the NTT module. This experience deepened my understanding of these critical areas and ignited an unyielding zeal to pursue them in graduate school.

Research Objectives At [redacted]. Modern cryptographic protocols, such as Homomorphic Encryption (HE) and ZKP, have tremendous potential in realizing new privacy-preserving applications; unfortunately, the wide-scale utility of these methods is severely limited by their high memory size, bandwidth, and computational overhead. These constructions typically involve polynomials of very large degrees ($\sim 2^{20}$) and hence require accelerated modules to operate on them. Furthermore, due to the lack of standardization of these constructions in terms of parameters (field size and polynomial degree), flexibility to support multiple parameters becomes a requirement. During my Ph.D., apart from only hardware implementations, I'd like to delve deeper into analyzing the underlying mathematical principles of these constructions and co-optimizing algorithms, protocols, and hardware to enable scalable computing on encrypted data.

User data is collected extensively by IoT end devices, which have minimal computing capabilities. These devices are typically battery-powered or energy-harvesting, making any complex operation an exacting task. Hence, energy efficiency is a crucial challenge when integrating modern cryptographic algorithms into these devices. The ubiquity of these devices further emphasizes the imperative of efficient, hardware-friendly protocols that guarantee security. In this area, my interest lies in implementing simplified, hardware-friendly protocols tailored to low-resource environments. Additionally, I look forward to designing secure integrated circuits that are resilient against side-channel and fault injection attacks.

Fit With Department. In graduate school, I am determined to develop expertise in designing special-purpose chips and efficient mapping of algorithms on FPGA. The ECE department at BU is my ideal choice because of the strong alignment of my interests with several projects undertaken by the department faculty. I am interested in working with [redacted] to develop privacy-preserving systems and HE accelerators. In particular, I would like to work on problems similar to [redacted]'s doctoral projects. I am also interested in studying the implementation of PQC constructions with [redacted]'s group. I eagerly look forward to working with [redacted]; his works on FPGA and GPU-based acceleration of algorithms across diverse fields fascinate me, and such an experience would be invaluable to broaden my horizons as a researcher. I would also be interested in learning more about [redacted]'s projects that involve chip design of lightweight cryptographic primitives for low-resource devices.

Future Plans & Career Goals. During my graduate studies, I aim to specialize in hardware design, focusing on cryptographic accelerators. After completing my Ph.D., I aim to pursue an industry research position to enable futuristic applications through hardware acceleration and simultaneously gain insights into the kind of problems the industry is dealing with. Reflecting on my past, I have been profoundly influenced by my advisors, who not only gave me project-specific advice but also shaped the way I think; hence, from a long-term perspective, I hope to pay forward this kind of positive influence as an academic.